JOURNAL OF FINANCIAL AND QUANTITATIVE ANALYSIS VOL. 31, NO. 4, DECEMBER 1996

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

The Creation and Resolution of Market Uncertainty: The Impact of Information Releases on Implied Volatility

Louis H. Ederington and Jae Ha Lee*

Abstract

We model and examine the impact of information releases on market uncertainty as measured by the implied standard deviation (ISD) from option markets. Distinguishing between scheduled and unscheduled announcements, we hypothesize that since the timing, although not the content, of scheduled announcements is known a priori, the pre-release ISD will impound the anticipated impact of important releases on price volatility and that the ISD will normally decline post-release as this uncertainty is resolved. Conversely, we hypothesize that the unexpected high volatility caused by major unscheduled releases will cause market participants to adjust upward their estimates of likely volatility over the remaining life of the option resulting in an increase in the ISD. Our evidence supports both hypotheses. The ISDs that we consider are from the T-Bond, Eurodollar, and Deutschemark options markets. We examine scheduled macroeconomic news releases such as the employment report and the PPI. We also find that the observed tendency for the ISD to fall on Fridays and rise on Mondays is due to the weekday pattern of scheduled news releases.

I. Introduction

We explore the impact of information releases on market participants' uncertainty regarding future security prices. While innumerable event studies have examined the impact of information disclosures on the first moment of investors' ex ante probability distributions regarding future prices, only a few, principally Patell and Wolfson (1979), (1981), have examined the impact on the second moment. Our measure of market uncertainty is the implied standard deviation (ISD) or implied variance (IMV) calculated from option prices.

We argue that a news announcement's impact on market uncertainty depends largely on whether the announcement is scheduled or unscheduled where, by *scheduled* we refer to news releases that the market knows beforehand are forthcoming

*Both authors, College of Business Administration, University of Oklahoma, Norman, OK 73019.
We acknowledge the helpful comments of Larry Benveniste, William Callahan, Edwin Elton, Craig Lewis, Scott Linn, and William Reinchenstein as well as Jennifer Conrad and Dan French (the referees). Seminar participants at the 1993 Financial Management Association meeting, the University of Oklahoma, the 1995 Southwest Finance Symposium, and the 1995 Utah Winter Finance Conference also provided useful suggestions. The Center for Financial Studies at the University of Oklahoma provided financial support for data acquisition. Lee was partially supported by a Noble Research Grant from the University of Oklahoma.

(such as earnings or dividend announcements or the consumer price index).[1] Since market participants know that important scheduled releases have the potential to cause large changes in the underlying security's price, we hypothesize that uncertainty, as measured by the ISD, will be high prior to their release. Following the release and accompanying price adjustment, the ISD should fall as this source of uncertainty is resolved. On the other hand, since their release is unanticipated, the high volatility due to important *unscheduled* announcements cannot be impounded in pre-release ISDs. Moreover, given the evidence on volatility clustering, it is likely that, after observing the unexpectedly high volatility induced by an unscheduled announcement, market participants will expect higher than normal volatility to continue. If so, they should revise upward their expectations regarding likely volatility over the remaining life of the option resulting in a rise in the ISD. Consequently, we hypothesize that the ISD tends to fall following important scheduled announcements but rise following important unscheduled announcements.

As in Ederington and Lee (1993), (1995), we examine the impact of macroeconomic information releases, such as the employment report and producer price index, on the T-Bond, Eurodollar, and Dollar/Deutschemark markets. While Ederington and Lee examine the impact of these announcements on futures prices, we explore the impact on the option market ISDs. Confirming our hypothesis, we find that, in the interest rate options markets, the ISD normally declines following the release of price impacting scheduled announcements. We also hypothesize and find that the decline tends to be greater the shorter the option's time-to-expiration and the greater the announcement's normal impact on actual volatility. As compared with the interest rate markets, we find that these announcements have little impact on Deutschemark futures prices and, consistent with this, little decline in the Deutschemark ISD following scheduled announcements is observed. Our evidence also indicates that, as hypothesized, the ISD normally rises following major unscheduled announcements.

Harvey and Whaley (1992) documents that the S&P 100's ISD tends to fall on Fridays and rise on Mondays. We observe the same pattern in the interest rate options markets. While they hypothesize that this pattern is due to buying/selling pressure as traders open positions on Monday and close them on Friday, our results indicate that in our markets, it is due to the tendency for scheduled announcements to be bunched on Fridays. Since interest rate ISDs tend to decline following major scheduled announcements, the question arises whether it is possible to exploit this predictability to earn abnormal returns. We find that it is not. Interestingly, transaction costs play only a minor role in our explanation. The major reason is that, even for delta neutral positions, losses due to the convexity of option prices and large price swings on the day of the announcement offset the profits due to the ISD decline.

The remainder of the paper is organized as follows. In Section II, we explore the impact of scheduled news announcements on market uncertainty. We model ISD behavior, review related studies, explain our data set, and test our

[1] Kim and Verrecchia (1991) uses the term "anticipated announcements" to refer to announcements whose occurrence is foreseen beforehand. Since the term "partially anticipated announcements" has been used to refer to situations in which content, not just timing, is anticipated, we choose to use the less ambiguous term "scheduled announcements."

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

hypotheses—including day-of-the-week and time-to-expiration effects. The impact of unscheduled announcements is explored in Section III. In Section IV, attention is turned to whether our results are consistent with market efficiency, i.e., whether it is possible to earn excess returns based on the tendency for the ISD to decline following scheduled announcements. Results are summarized in Section V.

II. The Impact of Scheduled Announcements on Market Uncertainty

A. A Model of ISD Behavior

Our measure of investors' uncertainty regarding future security prices is the implied standard deviation (ISD) calculated by Knight-Ridder Financial Inc. from daily option closing prices from 11/11/88 through 9/30/92 for T-Bonds, Eurodollars, and Deutschemarks. The fact that numerous option traders purchase the Knight-Ridder ISDs and use them in trading testifies to the market acceptance and perceived reliability of this volatility measure. The Knight-Ridder ISDs are an average of the ISDs calculated for the two nearest-the-money calls and the two nearest-the-money puts.[2] For calls, each of the ISDs is the value of σ_t obtained by substituting all other parameters into Black's (1976) options on futures model,

$$C_t = e^{-r_t T}\left[F_t N(d_{1t}) - E \cdot N(d_{2t})\right], \tag{1}$$

$$d_{1t} = \left[\ln\left(F_t/E\right) + 0.5\sigma_t^2 T_t\right]/\sigma_t\sqrt{T_t}, \tag{2}$$

$$d_{2t} = d_{1t} - \sigma_t\sqrt{T_t}, \tag{3}$$

where

C_t = current call price,
F_t = current futures price,
E = exercise price,
r_t = risk-free interest rate,
T_t = time to expiration of the option contract,
σ_t = standard deviation, and
$N(\cdot)$ = cumulative standard normal distribution.

The ISD calculated by solving these equations for σ_t is not a perfect measure of investors' volatility expectations. First, since the equation is nonlinear, the ISD will not be an unbiased measure of the market's true expectation, σ_t, (Butler and Schachter (1986)). More important, as discussed in Canina and Figlewski (1993), since the ISD is obtained by solving the above equations for observed option and futures prices, it will reflect any factors, such as liquidity, which affect option prices but are not incorporated in the Black model. In addition, price discreteness

[2]The Knight-Ridder ISDs are an equally weighted average. While they do not employ the weighting scheme of Latané and Rendleman (1976) or Chiras and Manaster (1978), since Knight-Ridder uses the two calls and two puts whose exercise prices are just above and just below the current price, the partial derivatives and elasticities of the four option prices, with respect to the standard deviation, are approximately the same. Consequently, the Latané and Rendleman and Chiras and Manaster weighting schemes would also imply roughly equal weights.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

and bid-ask spreads (Jorion (1995)) will introduce error. For simplicity, in our modeling, we interpret the ISD as reflecting investors' volatility expectations, but it should be kept in mind that it is not a precise measure.

In this paper, we posit that market participants do not expect volatility to be constant over time but instead anticipate that volatility will be higher on some days, specifically days with scheduled macroeconomic announcements, than others. If volatility is expected to vary, then, as shown in Merton (1973) and Hull and White (1987), σ_t^2 represents the mean anticipated daily volatility over the life of the option. Specifically,

$$\sigma_t^2 = T_t^{-1} \int_t^{te} \sigma^2(s)ds, \tag{4}$$

where $\sigma^2(s)$ is the instantaneous variance at time s. In discrete time, this may be approximated as

$$\sigma_t^2 = T_t^{-1} \sum_{u=t+1}^{te} \sigma_{u,t}^2, \tag{5}$$

where $\sigma_{u,t}^2$ is the anticipated variance on day u, given the information available on day t, and te is the option expiration date.[3] Consider the difference between σ_t^2 and σ_{t-1}^2. Since $T_{t-1} = T_t + 1$,

$$T_t\sigma_t^2 - T_t\sigma_{t-1}^2 - \sigma_{t-1}^2 = \sum_{u=t+1}^{te} \sigma_{u,t}^2 - \sum_{u=t+1}^{te} \sigma_{u,t-1}^2 - \sigma_{t,t-1}^2,$$

hence,

$$\sigma_t^2 - \sigma_{t-1}^2 = \left(\frac{1}{T_t}\right)\left[(\sigma_{t-1}^2 - \sigma_{t,t-1}^2) + \sum_{u=t+1}^{te} (\sigma_{u,t}^2 - \sigma_{u,t-1}^2)\right],$$

or, in percentage terms,

$$\frac{\sigma_t^2 - \sigma_{t-1}^2}{\sigma_{t-1}^2} = \left(\frac{1}{T_t}\right)\left[\frac{(\sigma_{t-1}^2 - \sigma_{t,t-1}^2)}{\sigma_{t-1}^2} + \sum_{u=t+1}^{te} \frac{(\sigma_{u,t}^2 - \sigma_{u,t-1}^2)}{\sigma_{t-1}^2}\right]. \tag{6}$$

As shown in equation (6), the change in the implied variance (IMV) is due to two effects: i) the removal of day t from the period covered by σ_{t-1}^2 (represented by the

[3]Volatilities, $\sigma_{u,t}^2$, may vary over time in either a deterministic or stochastic manner and equation (4) is used in both type models. However, Hull and White (1987), Ball and Roma (1994), and others show that if volatility is stochastic and the implied variance (IMV) is estimated by inverting equations (1)–(3), then the resulting IMV estimate will tend to underestimate average *expected* volatility over the life of the option. Fortunately, several studies indicate that this bias is small for at-the-money options as used here—less than 1 percent according to Heynen, Kemna, and Vorst (1994)—and all published studies of implied volatility to date use IMVs obtained by inverting equations (1)–(3) or similar models. We follow this procedure also and interpret IMV changes as generally reflecting changes in expected volatilities.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

first bracketed term in equation (6)), and ii) revisions in the expectations regarding volatilities on days $t + 1$ and later (the second bracketed term).

Consider the change in the IMV from the close on day $t - 1$ to the close on day t if a major scheduled announcement is released on day t. Since it is known on day $t - 1$ that the announcement will be released on day t, higher than normal volatility should be anticipated on day t. Hence, on day $t - 1$, the anticipated variance for day t will exceed the mean anticipated daily variance from day t through expiration, i.e., $\sigma^2_{t,t-1} > \sigma^2_{t-1}$ and the first bracketed term in equation (6) will be negative. In other words, the IMV will tend to decline because the period over which it is calculated will no longer include the (anticipated) high volatility day, t.

The change in the IMV will also incorporate revisions in expectations regarding volatilities on day $t + 1$ and after, as represented by the second bracketed term (the summation term) in equation (6). The term $(\sigma^2_{u,t} - \sigma^2_{u,t-1})$ represents the change from day $t - 1$ to day t in the market's expectation regarding volatility on the *same* future day u. According to rational expectations, the expectation as of day $t - 1$ should reflect all available information regarding likely volatility on day u. Consequently, $\sigma^2_{u,t}$ should only differ from $\sigma^2_{u,t-1}$ if new information is received on day t, and $(\sigma^2_{u,t} - \sigma^2_{u,t-1})$ should be uncorrelated with all information available at time $t - 1$, including whether an announcement is scheduled for t.

Suppose a major announcement of a given type, such as the CPI, is scheduled for day t and day u is a day soon after. While it is likely that higher than normal volatility will be observed on day t, rational expectations implies that, in general, the impact of a scheduled announcement on volatility on days t and u should already be reflected in the expectations formed on day $t - 1$. Depending on whether actual day t volatility is higher or lower than $\sigma^2_{t,t-1}$ following a scheduled announcement, expectations of likely volatility on day u may be revised upward or downward from $\sigma^2_{u,t-1}$. However, rational expectations imply that, since the normal impact of the announcement on day u volatility should already be reflected in $\sigma^2_{u,t-1}$, upward and downward revisions are equally likely and the mean revision across many such scheduled announcements should be approximately zero. Consequently, the mean change in IMV on days with scheduled announcements should be dominated by the first bracketed term and our first hypothesis is,

H1. The IMV will tend to fall on days with important scheduled announcements.

Conversely, if no major news release is scheduled on day t, but some are scheduled prior to expiration, then lower than normal volatility will normally be expected on day t, that is, $\sigma^2_{t,t-1} < \sigma^2_{t-1}$, so the first bracketed term in equation (6) will tend to be positive. Again, rational expectations imply that, across all days without scheduled announcements, the mean revision in expected volatilities on later days should be roughly zero, so again, the mean IMV change is dominated by the first bracketed term. This yields our second hypothesis,[4]

H2. The IMV will tend to rise on days with no scheduled announcements.

[4]The terms $\sigma^2_{t,t-1}$ and σ^2_{t-1} will partially reflect market participants' evaluations on day $t - 1$ of the likelihood of unscheduled announcements on day t and on day t through expiration, respectively. This hypothesis assumes that the likelihood of an unscheduled announcement on day t is viewed as no higher than that on the average day until expiration, which, given their unscheduled nature, seems reasonable. The impact of unscheduled announcements per se is considered below.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

In other words, the IMV will tend to rise on days with no scheduled announcements because a day with lower than normal expected volatility will be dropped from the period over which the IMV is calculated.

To this point, we have not differentiated between different types of scheduled announcements. However, our model implies that the decline in the IMV will be greater the higher the *anticipated* volatility on day t, $\sigma^2_{t,t-1}$, and this may vary across announcements. For instance, Ederington and Lee (1993) finds that the employment report has a much greater impact on actual volatility than other announcements. If market participants share this view, we should observe a sharper drop in the IMV following the release of the employment report than following other scheduled releases. Consequently, we hypothesize,

H3. The magnitude of the decline in the IMV following a scheduled announcement will be greater, the greater the *average* actual volatility associated with announcements of its type.

Note that to the extent high actual volatility on day t is anticipated, this hypothesis implies that actual volatility on days with scheduled announcements and the change in implied volatility over the remaining life of the option will be negatively correlated.

For the same bracketed term in equation (6), our model implies that the change in the implied variance is proportional to T_{t-1}. This is simply due to the fact that dropping one day from the period covered by σ^2_{t-1} has more impact on $(\sigma^2_t - \sigma^2_{t-1})$ if the time to expiration is short. Consequently, our fourth hypothesis is,

H4. The fall in the IMV on days with scheduled announcements, and the rise in the IMV on days without scheduled announcements will be greater, the shorter the time to expiration of the option.

B. Previous Evidence

In the two papers most closely related to ours, Patell and Wolfson (1979), (1981) examine the response of equity ISDs to yearly and quarterly earnings disclosures and find weak evidence of ISD declines following these disclosures. Similarly, in a paper developed independently of ours, Bonser-Neal and Tanner (1994) observes a decline in foreign exchange ISDs following several macroeconomic announcements. While only these papers hypothesize an ISD decline following scheduled announcements, a number of others have examined ISD behavior around announcements, including Cornell (1978), French and Frasier (1986), Bailey (1988), Franks and Schwartz (1991), Levy and Yoder (1993), and French, Swidler, and Trapani (1994).

A number of studies, including Latané and Rendleman (1976), Choi and Wohar (1992), Day and Lewis (1988), (1992), Lamoureux and Lastrapes (1993), Canina and Figlewski (1993), and Jorion (1995), examine the ability of the ISD to predict actual price volatilities. While Canina and Figlewski (1993) finds the ISD has very little predictive ability, most others find that the ISD has predictive ability but that it does not fully incorporate all available information including recent past volatilities. As noted above, this may be because of price discreteness and bid-ask

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

spreads, or because the ISD reflects any factor that influences the call price but is not in the Black formula. Additionally, Harvey and Whaley (1992) and Merville and Pieptea (1989) examine the ISD's predictability and behavior over time, while Engle and Mustafa (1992) compares its implications for volatility persistence with ARCH models.

C. The Markets and the ISD Measure

Our hypotheses concerning scheduled announcements are tested using daily data for T-Bond, Eurodollar, and Deutschemark futures and futures options contracts from 11/11/88 through 9/30/92. The former are the most heavily traded long-term and short-term interest rate futures contracts in the world, while the Dollar/Deutschemark contract is the most heavily traded exchange rate futures contract in the U.S. When these call options are exercised, one receives the underlying nearby futures contract. Since these futures contracts are close to expiration, they are close substitutes for the underlying cash market rates, so our ISDs should provide measures of uncertainty regarding future interest and exchange rates.[5] In all three markets, we study options for futures contracts maturing in March, June, September, or December. Strike prices are in increments of $1 per $100 par value (e.g., $101, $102, etc.) in the T-Bond market, in increments of $0.25 per $100 par value in the Eurodollar market, and in increments of $0.005 per mark in the Deutschemark market.

As noted above, the ISD estimates we analyze are based on those calculated and sold by Knight-Ridder Financial Inc. Since they are bought and widely used by traders, these Knight-Ridder ISDs should proxy market participants' consensus. We make one adjustment to the ISDs reported by Knight-Ridder. They use calendar, not market days, in their calculation. So, if Friday's ISD is calculated using T days, Monday's ISD is calculated using $T - 3$ days. As pointed out by French (1984), among others, this assumes that the variance of returns from Friday's close to Monday's close is three times the normal weekday close-to-close variance. Our evidence does not support this. Indeed, in the two interest rate markets, the three-day weekend return variance is actually less than the average weekday variance.[6] Consistent with this, Knight-Ridder's Monday ISDs exceed the preceding Friday's approximately 75 percent of the time for T-Bonds and approximately 60 percent for Eurodollars and Deutschemarks. Consequently, we adjust their ISDs to a market day basis. In particular, we use $\sigma = \sigma_{KR}\sqrt{T_c/T_m}$, where σ_{KR} is the ISD supplied by Knight-Ridder and T_m and T_c are market and calendar days to expiration. This follows French's (1984) recommendation that market days, T_m, be used in regard to σ but that calendar days, T_c, be used in the discount factor in equation (1). We use ISDs calculated for the nearby contract, switching to the next nearby contract one week before the nearby expires to min-

[5]The Eurodollar futures provides for cash settlement based on the three-month London Interbank Offer Rate (LIBOR). While this is an offshore (to avoid insurance fees and regulations) interbank rate, it is dollar denominated and highly correlated with purely domestic U.S. rates.

[6]This appears to be because few scheduled news releases occur on Mondays. When we examine only days with no scheduled announcements, the three-day weekend variance is approximately equal to the weekday variances. In the Deutschemark market, the three-day weekend variance is slightly larger than the average weekday but not as large as the calendar day assumption implies.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

imize any distortion caused by abnormal option price behavior as the expiration day approaches.

D. The Announcements and Their Impact on Prices

The scheduled news releases that we consider are listed in the Appendix. These are the 12 announcements that Ederington and Lee (1993) (hereafter, E&L) found to have a significant (0.02 level) impact on futures prices in the five-minute period just after release in at least one of the three markets.[7] Although its impact cannot be separated from day-of-the-week effects, we also consider below the possible impact of the weekly money supply announcement.

In order for there to be much uncertainty to resolve, an announcement must be viewed as likely to impact the price of the underlying security. This may not always be the case ex post. If the announced figure matches the market's expectation, then the futures price should be little changed. Nevertheless, if an announcement often (never) moves futures prices, anticipated volatility on the day it is released should be high (normal) and its release should (should not) affect the ISD. While E&L find that all 12 announcements impact returns in the five-minute interval just after the release in at least one market, we expect only announcements strong enough to impact daily volatility to matter in terms of market uncertainty. Consequently, in Table 1, we examine the impact on actual volatility. There, we report estimates of the E&L equation,

$$(7) \qquad |R_t - \bar{R}| = \alpha_0 + \sum_{k=1}^{K} \alpha_k D_{kt} + e_t,$$

where $R_t = \ln(F_t/F_{t-1})$, F_t = the futures closing price on day t, $\bar{R}$ = the mean of R_t over the sample period, and $D_{kt} = 1$, if announcement k is released on day t, and $D_{kt} = 0$ otherwise. Equation (7) is estimated using daily data from 11/11/88 through 9/30/92. As noted in E&L, if log returns are normally distributed with a constant mean but time-varying variance, $\sqrt{\pi/2}\,\alpha_0$ provides an estimate of the standard deviation of log returns on days with no scheduled announcements and $\sqrt{\pi/2}\,(\alpha_0 + \alpha_k)$ estimates the standard deviation on days when announcement k, and only k, is released. If announcement k impacts the market, $\alpha_k > 0$.

As in E&L, separate coefficients, α_k, cannot be estimated for all 12 announcements due to overlapping announcement dates. With a single exception, industrial production and capacity utilization are announced on the same day each month. We combine them as one announcement using the industrial production announcement date for the one month in question. In 42 of 46 months, construction spending and the NAPM survey are released on the same day. These two are combined as follows: $D_{kt} = 1$ if both are announced on day t, $D_{kt} = 0.5$ if only one of the two is

[7] All, except GNP, are monthly statistics. We do not consider quarterly and yearly announcements due to an inadequate number of data points. While GNP is a quarterly statistic, preliminary, revised, and final estimates of GNP are released in successive months. We include all three and treat it as monthly. The Bureau of Economic Analysis announced Gross National Product (GNP) statistics through December 1991 and Gross Domestic Product (GDP) beginning in January 1992. We use the term GNP for both.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

TABLE 1

Scheduled Announcements and Actual Volatility

Announcements (k)	Coefficients, α_k, ($\times 10^3$) with Standard Errors (in parentheses) T-Bond	Eurodollar	Deutschemark
Intercept (no announcement)	3.546* (0.142)	0.369** (0.021)	5.413** (0.203)
Construction Spending/ NAPM Survey	1.046* (0.552)	0.148* (0.083)	0.229 (0.792)
Consumer Price Index	2.014** (0.542)	0.189** (0.081)	0.017 (0.777)
Durable Goods Orders	0.355 (0.533)	0.178* (0.080)	−0.064 (0.764)
Employment	4.147** (0.539)	0.769** (0.081)	2.020** (0.773)
Federal Budget	0.437 (0.533)	0.130 (0.080)	1.791** (0.764)
Gross National Product	0.659 (0.538)	0.125 (0.080)	0.268 (0.772)
Industrial Production/ Capacity Utilization	−0.247 (0.572)	0.007 (0.085)	−1.098 (0.811)
Merchandise Trade Deficit	0.180 (0.558)	0.115 (0.083)	2.905** (0.800)
Producer Price Index	3.077** (0.578)	0.393** (0.086)	1.807* (0.826)
Retail Sales	0.290 (0.568)	−0.027 (0.084)	0.753 (0.806)
Adjusted R^2	0.085	0.100	0.019

Estimates of the equation,

$$|R_t - \bar{R}| = \alpha_0 + \sum_{k=1}^{K} \alpha_k D_{kt} + e_t,$$

using daily data from 11/11/88 through 9/30/92 are reported. $R_t = \ln(F_t/F_{t-1})$, F_t = the closing futures price on day t, D_{kt} = 1 if announcement k is announced on day t and D_{kt} = 0 otherwise. * and ** denote significance at the 0.05 and 0.01 levels in one-tailed tests, respectively.

announced on day t, and $D_{kt} = 0$ otherwise.[8] No other announcement pairs have more than 16 days in common.

Consider first the interest rate market results in Table 1. As in E&L, the employment report appears to have the greatest impact followed by the PPI. As measured by $[(\alpha_0+\alpha_k)/\alpha_0]^2$, the results in Table 1 imply that the variance of returns in the T-Bond market on days when the employment report is released is 4.7 times the variance on days with no scheduled announcements. In the Eurodollar market, this ratio is 9.5. For the PPI release, the ratios are 3.5 for T-Bonds and 4.3 for Eurodollars. The CPI is also significant at the 0.01 level in both markets. Most

[8]Defining this dummy as 0.5 when the announcements occur separately assumes each announcement has half the effect of the two together and is admittedly arbitrary. Since these two announcements are issued on different days in only four months, there is no way we can disentangle their separate effects. Fortunately, since there are only four such cases, how the dummies are defined makes little difference. To see whether our results are sensitive to this choice, we reran the three regressions in Tables 1 and 3 without dummies for these announcements in these four months. With the exception of the Eurodollar results in Table 3 where the coefficient was changed from −1.249 to −0.850, the coefficients were little changed and no significance results were changed.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

of the other announcements have positive coefficients and some are significant at the 0.05 level. However, their coefficients imply only a small increase in the variance on release dates so it is doubtful if much uncertainty is resolved by their release. Consequently, in our interest rate ISD analysis, we consider the impact of the employment report and PPI separately and the other 10 announcements as a group.

The impact of the announcements in the Deutschemark market is much weaker. While the coefficients for the merchandise trade deficit, the employment report, and the federal budget are significant at the 0.01 level, the adjusted R^2 is quite small. Consequently, we expect the impact of these announcements on the Deutschemark ISDs to be slight.

E. Results: The Impact of Scheduled Announcements

Evidence on H1 and H2 is shown in Table 2 where we report statistics on the log percentage change in σ_t, where σ_t is the ISD calculated from closing options and futures prices on day t.[9] While our model in Section II.A was expressed in terms of IMV for simplicity, we choose to analyze the log percent change in the ISD empirically since that has been the practice in previous studies.

In the interest rate options markets, H1 and H2 are both clearly confirmed. In the T-Bond market, the ISD rises an average of 0.505 percent on days with no scheduled announcements and falls an average of 0.781 percent on days with at least one.[10] These two means are significantly different from zero as well as each other at the 0.01 level and the same is true of the medians. The ISD declines on 59.0 percent of the announcement days but on only 42.5 percent of days without scheduled announcements; both proportions are significantly different from 0.5 at the 0.01 level. Roughly the same, but somewhat stronger results are observed in the Eurodollar market where the ISD rises, on average, 0.721 percent on days without scheduled announcements and falls 1.398 percent on days with scheduled announcements. While these percentage changes in the ISD of 1.5 percent or less may appear small, for $T = 38$, the median in our sample, they imply that the market expects that the volatility on days when a scheduled announcement is released, $\sigma^2_{t,t-1}$, will be more than double the volatility on days without scheduled

[9]All of our announcements are released well before the markets close. With the exception of the federal budget announcement, which is released at 2:00 p.m. (Eastern Time), all are released in the morning and most are released at 8:30 a.m. shortly after the markets open. Hence, their impact should be fully reflected in closing options and futures prices from which σ_t is calculated.

[10]Hypotheses H1 and H2 were developed in terms of expected volatilities. As noted in footnote 3, Hull and White (1987), Ball and Roma (1994), and other studies show that if anticipated volatility is stochastic, the IMV will tend to slightly underestimate the true average expected volatility over the life of the option. Consequently, we consider whether fluctuations in this bias could account for the results in Table 2. Ball and Roma (1994) shows that the bias depends on 1) the variance of the volatilities and 2) whether the option is out-, in-, or at-the-money. Since the latter is approximately constant (at-the-money) in our data, to account for the pattern in Table 2 according to Ball and Roma, var(σ^2) would have to rise following scheduled announcements and fall on other days. We can think of no reason to expect such a pattern. More important, models of the bias, e.g., Heynen, Kemna, and Vorst (1994), indicate it is too small to account for the results in Table 2—particularly the ISD drop following release of the employment report.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

TABLE 2

Impact of Scheduled Announcements on Implied Volatility

Panel A. T-Bond

Statistics	Days with Sch. Ann. (393 days)	Days without Sch. Ann. (588 days)	Employment Ann. (46 days)	PPI Ann. (45 days)	Other Ann. (302 days)
Mean (μ) of $\ln(\sigma_t/\sigma_{t-1})$	−0.781	0.505	−4.406	−1.111	−0.179
t-Statistic (H_0 : $\mu = 0$)	−3.357**	3.113**	−4.723**	−1.326	−0.813
t-Statistic (H_0 : $\mu_{SA} = \mu_{NSA}$)	−4.534**	NA	−5.187**	−1.893	−2.499*
Median (M) of $\ln(\sigma_t/\sigma_{t-1})$	−0.715	0.465	−4.504	−0.840	−0.420
Signed Rank Statistic (H_0 : $M = 0$)	−10728**	15355**	−384**	−117	−4001**
KW Statistic (H_0 : $M_{SA} = M_{NSA}$)	37.953**	NA	49.099**	5.958*	18.271**
Percentage (P) of Days $\ln(\sigma_t/\sigma_{t-1}) < 0$	59.0%	42.5%	82.6%	57.8%	55.6%
z-Statistic (H_0 : $P = 0.5$)	3.581**	−3.629**	4.423*	1.043	1.956
χ^2-Statistic (H_0 : $P_{SA} = P_{NSA}$)	25.710**	NA	27.661**	3.959*	13.772**
Standard Deviation of $\ln(\sigma_t/\sigma_{t-1})$	4.612	3.932	6.327	5.621	3.835

Panel B. Eurodollar

Mean (μ) of $\ln(\sigma_t/\sigma_{t-1})$	−1.398	0.721	−7.408	−0.889	−0.559
t-Statistic (H_0 : $\mu = 0$)	−4.398**	3.262**	−6.300**	−1.069	−1.733
t-Statistic (H_0 : $\mu_{SA} = \mu_{NSA}$)	−5.474**	NA	−6.794**	−1.870	−3.273**
Median (M) of $\ln(\sigma_t/\sigma_{t-1})$	−0.883	0.837	−4.964	−0.930	−0.405
Signed Rank Statistic (H_0 : $M = 0$)	−9382**	11792**	−446**	−79	−3128*
KW Statistic (H_0 : $M_{SA} = M_{NSA}$)	25.732**	NA	44.447**	3.203	10.862**
Percentage (P) of Days $\ln(\sigma_t/\sigma_{t-1}) < 0$	56.0%	43.5%	80.4%	53.3%	52.7%
z-Statistic (H_0 : $P = 0.5$)	2.371*	−3.134**	4.128**	0.447	0.921
χ^2-Statistic (H_0 : $P_{SA} = P_{NSA}$)	14.599**	NA	23.366**	1.626	6.656**
Standard Deviation of $\ln(\sigma_t/\sigma_{t-1})$	6.302	5.358	7.976	5.579	5.603

Panel C. Deutschemark

Statistics	Days with Sch. Ann. (393 days)	Days without Sch. Ann. (588 days)	Employment Ann. (46 days)	Trade Deficit Ann. (47 days)	Other Ann. (300 days)
Mean (μ) of $\ln(\sigma_t/\sigma_{t-1})$	0.232	−0.256	−0.309	−0.478	0.426
t-Statistic (H_0 : $\mu = 0$)	1.138	−1.623	−0.621	−0.682	1.845
t-Statistic (H_0 : $\mu_{SA} = \mu_{NSA}$)	1.893*	NA	−0.101	−0.310	2.439*
Median (M) of $\ln(\sigma_t/\sigma_{t-1})$	−0.020	−0.266	−0.220	0.241	−0.000
Signed Rank Statistic (H_0 : $M = 0$)	657	−9758*	−84	−17	1120
KW Statistic (H_0 : $M_{SA} = M_{NSA}$)	3.052*	NA	0.003	0.201	3.878*
Percentage (P) of Days $\ln(\sigma_t/\sigma_{t-1}) < 0$	50.4%	54.1%	56.5%	46.8%	50.0%
z-Statistic (H_0 : $P = 0.5$)	0.151	1.979*	0.885	−0.438	0.000
χ^2-Statistic (H_0 : $P_{SA} = P_{NSA}$)	1.293	NA	0.102	0.926	1.328
Standard Deviation of $\ln(\sigma_t/\sigma_{t-1})$	4.044	3.820	3.373	4.810	4.001

Statistics are reported for $\ln(\sigma_t/\sigma_{t-1})(\times 10^2)$, where σ_t is the ISD calculated from daily closing options and futures prices by Knight-Ridder Financial Inc. adjusted to a market day basis. Nonannouncement days are defined as days with none of our scheduled macroeconomic announcements. Daily data from 11/11/88 through 9/30/92 are utilized. * and ** denote significance at the 0.05 and 0.01 levels, respectively, in two-tailed tests. NA denotes not available. KW denotes the Kruskal-Wallis test statistic for the null hypothesis that the medians are equal.

announcements in the T-Bond market and more than four times volatility on days without scheduled announcements in the Eurodollar market.[11]

Consistent with the results in Table 1, it is clear that the employment report has a particularly strong impact. On days when it is released, the ISD declines an average of 4.4 percent in the T-Bond market and 7.4 percent in the Eurodollar

[11] According to the results in Table 2, $\ln(\sigma_t/\sigma_{t-1})$ averages −0.00781 in the T-Bond market, implying $\sigma_t^2 = 0.9845\sigma_{t-1}^2$. Substituting this and $T = 38$ into equation (6) and imposing the rational expectations condition that the summation term averages zero, yields $\sigma_{t,t-1}^2(\text{SA}) = \sigma_{t-1}^2 + 38(0.0115)\sigma_{t-1}^2 = 1.437\sigma_{t-1}^2$, where the SA denotes that a scheduled announcement occurs on day t. Similarly, the mean ISD change of +0.00505 on days without scheduled announcements implies $\sigma_{t,t-1}^2(\text{NSA}) = 0.614\sigma_{t-1}^2$. Hence, the implication is that, if an announcement is scheduled for day t, the anticipated variance, $\sigma_{t,t-1}^2(\text{SA}) = (1.437/0.614)\sigma_{t,t-1}^2(\text{NSA}) = 2.34\sigma_{t,t-1}^2(\text{NSA})$, the anticipated variance on days without scheduled announcements. When the same procedure is repeated using the Eurodollar means from Table 2, $\sigma_{t,t-1}^2(\text{SA}) = 4.57\sigma_{t,t-1}^2(\text{NSA})$.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

market. For $T = 38$, these figures imply that market participants anticipate that the volatility on days when the employment report is released will be 6.8 (13.9) times the normal variance on days without scheduled announcements in the T-Bond (Eurodollar) market. These are roughly consistent with, but somewhat higher than, the ratios for actual volatilities implied by the results in Table 1: 4.7 times normal volatility on days without scheduled announcements for T-Bonds and 9.5 for Eurodollars as discussed above. The ISD declines following more than 80 percent of the employment releases in both markets. On days when one of the 10 announcements other than the employment report or PPI is released, the decline in the ISD is considerably less, 0.18 percent (T-Bond) and 0.56 percent (Eurodollar).

Since, in Table 1, we found that scheduled announcements have little impact on actual volatility in the Deutschemark market, we expect little uncertainty to be resolved there by their release. Consistent with this, we find little evidence in Panel C of Table 2 of uncertainty resolution. The mean of $\ln(\sigma_t/\sigma_{t-1})$ is positive on announcement days and negative on days without scheduled announcements—the opposite of our hypothesis—and neither is significantly different from zero. The means and medians for the employment report and trade deficit have the hypothesized signs but are small and insignificant.

In H3, we hypothesized that the decline in the ISD following a scheduled announcement is greater, the greater the usual day t volatility associated with announcements of that type and, therefore, the greater the expected day t volatility. Certainly, our employment report results confirm this. Further evidence is presented in Table 3. There, we present estimates of the equation (labeled Model 1),

$$\ln\left(\sigma_t/\sigma_{t-1}\right) = \beta_0 + \sum_{k=1}^{K} \beta_k D_{kt} + u_t, \tag{8}$$

where D_{kt} is the dummy variable previously defined to identify days on which announcement k is released. β_0 provides an estimate of $\ln(\sigma_t/\sigma_{t-1})$ on days with no scheduled announcements and $\beta_0 + \beta_k$ provides an estimate for days when k, and only k, is released. H2 implies $\beta_0 > 0$, H1 implies $\beta_0 + \beta_k < 0$ for all k, and H3 implies that the β_k from Table 3 and the α_k from Table 1 should be negatively correlated.

Residuals from OLS estimates of equation (8) are negatively autocorrelated. Our interpretation of this phenomenon is that, due to transaction costs, the discreteness of option and futures prices, and the influence of other factors on options prices, ISDs calculated from option and futures prices measure the market's true uncertainty with error.[12] Since $\ln(\sigma_t/\sigma_{t-1})$ and $\ln(\sigma_{t-1}/\sigma_{t-2})$ share the term σ_{t-1}, any measurement error will result in negative serial correlation. Based on ARIMA diagnostics, equation (8) is estimated using an AR(3) model.[13]

[12]Harvey and Whaley (1992) has also observed that daily S&P 100 ISD changes are negatively correlated. They hypothesize that this is due to a tendency for the S&P 100 index to lag the options market. In other words, if good (bad) news arrives late in the trading day, it raises (lowers) option prices, but individual stock prices are slower to adjust so the calculated ISD rises (falls). The next day, the ISD falls (rises) as the individual stocks adjust. That explanation seems less credible here since Ederington and Lee (1993) documents that prices in these markets finish adjusting within minutes.

[13]The coefficients of the three lagged error terms are −0.1728, −0.0744, and −0.1260, respectively, in the T-Bond market, and −0.0875, −0.0999, and −0.0943 in the Eurodollar market.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

TABLE 3

Impact of Individual Scheduled Announcements on Implied Volatility: Regression Evidence

	Coefficients, β_k, with Standard Errors (in parentheses)					
	Model 1 ($\times 10^2$)			Model 2 ($\times 10$)		
Announcements (k)	T-Bond	Eurodollar	Deutschemark	T-Bond	Eurodollar	Deutschemark
Intercept (no announcement)	0.568** (0.138)	0.640** (0.195)	−0.243 (0.148)	2.785** (0.540)	3.266** (0.753)	−1.084* (0.573)
Construction Spending/ NAPM Survey	−0.788 (0.637)	−1.249 (0.871)	0.790 (0.612)	−2.472 (2.639)	−8.664** (2.891)	4.687 (2.048)
Consumer Price Index	−2.201** (0.629)	−2.319** (0.860)	0.090 (0.602)	−6.256** (2.448)	−12.066** (4.910)	1.295 (2.620)
Durable Goods Orders	0.083 (0.616)	0.651 (0.842)	0.470 (0.591)	3.697 (3.334)	6.773* (3.710)	2.944 (1.797)
Employment	−4.598** (0.625)	−7.957** (0.855)	0.012 (0.599)	−19.053** (2.174)	−18.206** (1.989)	−1.346 (2.161)
Federal Budget	−0.758 (0.614)	−0.603 (0.841)	−0.064 (0.591)	−4.799 (3.652)	−1.584 (4.217)	0.702 (2.182)
Gross National Product	−0.928 (0.619)	−0.590 (0.848)	0.677 (0.596)	−4.341 (3.124)	0.280 (3.197)	2.640 (1.654)
Industrial Production/ Capacity Utilization	−0.530 (0.679)	−0.709 (0.919)	0.874 (0.634)	−1.577 (2.137)	−4.867 (5.447)	4.287 (2.980)
Merchandise Trade Deficit	−0.616 (0.654)	−1.351 (0.893)	−0.575 (0.621)	1.740 (2.485)	−4.842 (4.908)	−2.883 (2.630)
Producer Price Index	−1.274* (0.677)	−0.752 (0.922)	0.414 (0.643)	−6.148** (1.668)	−9.873* (4.451)	2.305 (3.358)
Retail Sales	−0.705 (0.669)	−1.417 (0.903)	1.163 (0.627)	−4.642** (1.985)	−4.807 (5.656)	7.307 (3.186)
Total R^2	0.113	0.118	0.018	0.136	0.124	0.026

Estimates of two regressions are reported,

Model 1: $\ln(\sigma_t/\sigma_{t-1}) = \beta_0 + \sum_{k=1}^{K} \beta_k D_{kt} + u_t$, where $u_t = \sum_{i=1}^{3} \alpha_i u_{t-i} + \epsilon_t$.

Model 2: $\ln(\sigma_t/\sigma_{t-1}) = (1/2T_t)[\beta_0 + \sum_{k=1}^{K} \beta_k D_{kt}] + v_t$, where $v_t = \sum_{i=1}^{3} \alpha_i v_{t-i} + \epsilon_t$.

where

σ_t = the implied standard deviation calculated from daily closing option and futures prices as reported by Knight Ridder Financial Inc. and adjusted to a market day basis,

D_{kt} = 1 if announcement k is released on day t, D_{kt} = 0 otherwise, and

T_t = the number of market days to expiration of the option contract.

Daily data from 11/11/88 through 9/30/92 are utilized. * and ** denote significance at the 0.05 and 0.01 levels in one-tailed tests, respectively.

As shown in Table 3, in the two interest rate markets, β_0 is positive and significant at the 0.01 level, further confirming H2. With the exception of durable goods orders, all β_k have the hypothesized negative sign (H1). Consistent with H3, we find that the employment report has a much greater impact on the interest rate ISDs than any of the other reports. The PPI is only significant in the T-Bond market (at the 0.05 level), while the CPI announcement is significant at the 0.01 level in both markets. Further confirming H3, the correlation coefficient between the β_k from Table 3 and the α_k from Table 1 is −0.860 for T-Bonds and −0.805 for Eurodollars. Both are significant at the 0.01 level. If the employment report is removed from the sample, the correlation is still significant in the T-Bond market (0.05 level), but not in the Eurodollar market.

The results for Deutschemarks mirror those in Table 2, i.e., most coefficients are insignificant and many have the wrong sign. Of course, this is consistent with the findings in Table 1 that the announcements have little impact on the underlying security's price in this market so there is little uncertainty surrounding

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

these announcements to be resolved. Having established that the impact of the announcements on the Deutschemark ISD is inconsequential, the remainder of the paper focuses solely on the interest rate markets.

F. Results: Time to Expiration

In H4, we hypothesized that the ISD tends to fall more on days with scheduled announcements and rise more on days without scheduled announcements, the closer the option is to expiration. Evidence on this hypothesis is presented in Table 4, where means and medians of $\ln(\sigma_t/\sigma_{t-1})$ are stratified by market days to expiration: 5–25, 26–46, and 46+ days. As hypothesized, for days with no scheduled announcements, the mean of $\ln(\sigma_t/\sigma_{t-1})$ is greatest for the subset of options with between five and 25 days to expiration averaging 0.82 percent (T-Bonds) and 1.94 percent (Eurodollars). However, while the relative ranking of the 26–46 and 46+ groups in the T-Bond market is consistent with our hypothesis, it is not in the Eurodollar market. The last employment report to be released prior to expiration clearly has a greater impact than earlier reports, −7.4 percent T-Bonds and −10.3 percent Eurodollars but, again, the relative ranking of the two earlier reports (26–46 vs. 46+) is unclear. No consistent pattern is observed for the PPI or "all other announcements."

TABLE 4

Implied Volatility Changes and Time-to-Expiration

Number of Days to Expiration		Days with Sch. Ann.	Days without Sch. Ann.	Employment Ann.	PPI Ann.	Other Ann.
Panel A. T-Bond						
5–25	(1)	−0.750**	0.821	−7.351**	−2.405	0.626
		(−0.715)**	(0.894)	(−9.535)**	(−0.650)*	(−0.066)
26–46	(2)	−0.797**	0.500	−2.918**	0.084	−0.607**
		(−0.618)**	(0.429)	(−4.339)**	(−0.227)	(−0.560)**
46+	(3)	−0.796**	0.222	−3.041**	−0.315	−0.508**
		(−0.824)**	(0.272)	(−2.757)**	(−1.375)	(−0.623)**
F-Statistic ($H_0: \mu_1 = \mu_2 = \mu_3$)		0.004	1.130	2.582	0.968	3.170*
KW Statistic ($H_0: M_1 = M_2 = M_3$)		(0.147)	(3.222)	(6.367)*	(0.686)	(2.195)
Panel B. Eurodollar						
5–25	(1)	−1.639**	1.938	−10.271**	−0.484	−0.292*
		(−1.354)**	(1.970)	(−13.729)**	(−0.305)	(−0.544)**
26–46	(2)	−1.381**	−0.267	−4.394*	−2.725	−0.757
		(−1.639)*	(0.087)	(−3.670)*	(−2.351)	(−0.695)
46+	(3)	−1.149**	0.517	−6.322**	0.394	−0.661*
		(−0.517)**	(0.678)	(−4.404)**	(0.374)	(−0.291)*
F-Statistic ($H_0: \mu_1 = \mu_2 = \mu_3$)		0.206	8.635**	2.427	1.208	0.197
KW Statistic ($H_0: M_1 = M_2 = M_3$)		(0.315)	(17.247)**	(3.222)	(2.332)	(0.143)

Mean and median (in parentheses) values of the percentage change in the ISD, $\ln(\sigma_t/\sigma_{t-1})(\times 10^2)$, are reported where the sample is stratified by the number of market days until the option contract expires. Daily data from 11/11/88 through 9/30/92 are utilized. * and ** on the mean and median in the announcement column denote rejection, at the 0.05 and 0.01 levels, respectively, of the null hypothesis that the mean or median for this group equals the mean or median on nonannouncement days. KW denotes the Kruskal-Wallis test statistic for the null hypothesis that the medians are equal.

According to equation (6), for the same bracketed term, the percentage change in the IMV is proportional to $(1/T_t)$ where T_t is the time-to-expiration, which

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

implies that the percentage change in the ISD should be approximately proportional to $1/(2T_t)$. This implies that the regression model in Table 3 is more appropriately expressed as,

$$(9) \qquad \ln\left(\sigma_t/\sigma_{t-1}\right) = \left(\frac{1}{2T_t}\right)\left[\beta_0 + \sum_{k=1}^{K}\beta_k D_{kt}\right] + v_t.$$

Estimates of equation (9) are reported as Model 2 in Table 3. In general, the results are quite similar to those for Model 1 (note that the coefficients are not comparable). However, Model 2's adjusted R^2 is somewhat higher, particularly for T-Bonds.

G. Results: Day-of-the-Week Patterns

Harvey and Whaley (1992) documents a day-of-the-week pattern in the S&P 100 option market ISD. In particular, they observe that the ISD tends to decline on Fridays and rise on Mondays (or more precisely from Friday close to Monday close). They attribute this pattern to buying/selling pressure, i.e., that the ISD tends to fall on Fridays as traders close positions prior to the weekend and that it tends to rise on Mondays as they reopen these positions.

Our results suggest an alternative explanation. As shown in the Appendix, scheduled announcements are not spread evenly over the week. The employment report is almost always issued on Friday, and the PPI is normally issued on Thursday or Friday. Monday is the slowest announcement day. Since we have documented that the ISD tends to fall following scheduled announcements and rise on days without scheduled announcements, that could conceivably explain the pattern. In addition, the weekly money supply figures are released after the markets close on Thursday and would, therefore, be reflected in the Friday ISD. For the reasons described above, the day-of-the-week results—Monday in particular—are quite sensitive to whether the ISD is calculated using calendar days, as Knight-Ridder does, or market days, which we view as more appropriate. Consequently, we present results for both in Table 5.

Consider Mondays first. When the ISD is calculated using calendar days, the ISD shows a clear tendency to increase from Friday close to Monday close in both markets. When, however, the ISD is recalculated using market days, this Monday effect disappears. Indeed, a tendency for the ISD to decline from Friday close to Monday close is apparent in the Eurodollar market. In both markets, there is a (significant) tendency for the ISD to decline on Fridays, whether the ISD is calculated using calendar or market days. The evidence in Table 5 indicates that this is due to the fact that important scheduled announcements are bunched on Friday, since there is a strong tendency for the ISD to decline on Fridays with scheduled announcements but not on Fridays without scheduled announcements. On the 45 Fridays when the employment report is released, the ISD (calculated using market days) declines an average of 4.40 percent in the T-Bond market; on the 88 Fridays when a scheduled announcement other than the employment report is released, the ISD declines 0.50 percent; and, on the 64 Fridays without a scheduled announcement, the ISD rises 0.02 percent. Only the first is significantly different

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

TABLE 5

Day-of-the-Week Effects on Implied Volatility

Panel A. T-Bond

Weekdays	Calendar Days			Market Days		
	All Days	Days with Sch. Ann.	Days without Sch. Ann.	All Days	Days with Sch. Ann.	Days without Sch. Ann.
Monday (M)	2.186 (1.510)	1.396 (1.505)	2.356 (1.549)	0.044 (−0.126)	−0.480 (0.012)	0.156 (−0.129)
Tuesday (T)	0.161 (−0.049)	−0.895** (−1.043)**	0.768 (0.207)	0.429 (0.212)	−0.628** (−0.667)**	1.037 (0.592)
Wednesday (W)	−0.056 (0.148)	−0.581 (−0.702)**	0.279 (0.467)	0.402 (0.549)	−0.181 (−0.405)**	0.774 (0.936)
Thursday (R)	−0.231 (0.000)	−0.349 (−0.776)	−0.157 (0.321)	0.284 (0.356)	0.140 (−0.370)	0.375 (0.732)
Friday (F)	−1.796 (−1.435)	−2.447** (−2.316)**	−0.443 (−0.439)	−1.221 (−0.828)	−1.818** (−1.657)**	0.019 (−0.112)
F-Statistic ($H_0: \mu_M = \mu_T = \mu_W = \mu_R = \mu_F$)	20.490**	5.927**	9.302**	5.309**	2.878*	1.308
t-Statistic ($H_0: \mu_M = \mu_{T-R}$)	6.195**	3.283**	4.836**	−0.988	−0.448	−1.472
t-Statistic ($H_0: \mu_F = \mu_{T-R}$)	−4.379**	−3.226**	−1.856	−4.079**	−2.863**	−1.843*
Panel B. Eurodollar						
Monday (M)	1.376 (0.938)	0.184 (0.938)	1.631 (1.005)	−0.061 (−0.677)	−2.400 (−1.209)	−0.215 (−0.610)
Tuesday (T)	0.396 (0.374)	−0.464 (−1.175)	0.890 (0.666)	0.379 (0.424)	−0.374 (−0.758)	0.812 (0.816)
Wednesday (W)	−0.007 (0.575)	−0.731 (−0.879)*	0.456 (0.702)	0.491 (0.878)	−0.219 (−0.430)*	0.944 (1.049)
Thursday (R)	0.196 (0.617)	−1.361** (−0.718)**	1.173 (1.170)	0.738 (0.864)	−0.915** (−0.458)**	1.777 (1.554)
Friday (F)	−2.272 (−1.253)	−3.281** (−2.274)**	−0.174 (0.564)	−1.689 (−0.830)	−2.680** (−1.950)**	0.370 (0.913)
F-Statistic ($H_0: \mu_M = \mu_T = \mu_W = \mu_R = \mu_F$)	10.305**	3.977**	1.640	5.956**	2.917*	2.506*
t-Statistic ($H_0: \mu_M = \mu_{T-R}$)	2.638**	1.049	1.552	−2.519*	−1.721	−2.759**
t-Statistic ($H_0: \mu_F = \mu_{T-R}$)	−4.524**	−3.206**	−1.538	−4.152**	−2.944**	−1.239

Mean and median (in parentheses) values of the percentage change in the ISD, $\ln(\sigma_t/\sigma_{t-1})(\times 10^2)$, are reported where the sample is stratified by day-of-the-week. Daily data from 11/11/88 through 9/30/92 are utilized. * and ** on the mean and median in the announcement column denote rejection, at the 0.05 and 0.01 levels, respectively, of the null hypothesis that the mean or median for this group equals the mean or median on nonannouncement days.

from zero at the 0.01 level, so most of the decline on announcement Fridays is due to the employment report. The same pattern is observed in the Eurodollar market where the mean ISD changes are −7.55 percent, −0.19 percent, and +0.37 percent for the same three Friday groups, respectively. If either selling pressure or weekly money supply announcements were responsible for the Friday pattern, we should observe the same pattern on Fridays without scheduled announcements as well. It is interesting that the average rise in the ISD on Fridays without scheduled announcements is somewhat less than the rise on most other days without scheduled announcements (market day calculations). Perhaps the money supply announcements or selling pressure have a minor effect on implied volatility.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

III. The Impact of Unscheduled Announcements on Market Uncertainty

A. A Model of ISD Behavior following Unscheduled Announcements

Attention is now turned from scheduled to unscheduled announcements. Consider the first bracketed term in equation (6). While major unscheduled announcements will cause actual volatility to be higher than normal on day t, this high volatility will not be anticipated on day $t - 1$ since the announcement is unscheduled, i.e., a surprise. In general, therefore, $\sigma^2_{t,t-1}$, anticipated volatility on day t as of day $t - 1$, should vary randomly around σ^2_{t-1}, anticipated mean daily volatility over the life of the option, and the first bracketed term in equation (6) should average approximately zero over days with major unscheduled announcements.

Consequently, the impact of unscheduled announcements on the IMV or ISD depends on whether the announcement, and the accompanying higher than expected volatility on day t, cause market participants to change their expectations regarding future volatilities over the remaining life of the option (the second bracketed term in equation (6)). In the case of scheduled announcements, we argued that $(\sigma^2_{u,t} - \sigma^2_{u,t-1})$ should vary randomly around zero for $u > t$ since, on average, the impact of the announcement on day u volatility should be anticipated on day $t - 1$. This will not be the case for major unscheduled announcements since their release is unanticipated as of day $t - 1$. As argued by Brown, Harlow, and Tinic (1993), one can think of situations in which anticipated future volatilities would be revised upward and situations in which they would be revised downward following unscheduled announcements. For instance, an unscheduled announcement or rumor that a company is a takeover target would probably be followed (and be expected to be followed) by a period of high uncertainty and volatility due to possible news and rumors of offers and counteroffers. Conversely, a subsequent unscheduled announcement that management is accepting a takeover offer might reduce anticipated future volatility. Following the surprise announcement of the death or resignation of the Fed Chairman, the market will likely anticipate higher than normal volatility as rumors of successors follow. Conversely, IMVs will likely be revised downward following the announcement of the appointment of that successor.

Many studies, ARCH-GARCH studies in particular, have found evidence of volatility clustering or persistence, i.e., high (low) volatility on day t tends to be followed by high (low) volatility on day $t + 1$.[14] As Engle and Mustafa (1992) and others argue, if market participants are rational, this volatility clustering effect should be impounded in their expectations. In particular, if actual volatility is higher (lower) than expected on day t, market participants should revise upward (downward) their expectations of volatilities on days $t + 1$ and later. Consistent with this, Engle and Mustafa (1992) finds evidence of an ARCH process in ISDs and studies of the term structure of volatility, Heynen, Kemna, and Vorst (1994), and Xu and Taylor (1995), find that short- and long-term expected volatilities tend to be revised together. Based on this evidence, we hypothesize:

[14] For a review of the ARCH-GARCH evidence see Bollerslev, Chou, and Kroner (1992).

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

H5. If actual day t volatility is higher (lower) than expected, market participants will revise their expectations of likely volatilities on days $t + 1$ and later upward (downward) causing IMVs to rise (fall).

Note that while hypotheses H1 and H2 referred to mean behavior of the IMV on days with and without scheduled announcements as groups, H5 pertains to IMV behavior on individual days and applies to both days with and without scheduled announcements.

Since, unlike scheduled announcements, unscheduled announcements will generally cause actual volatility to be higher than expected on day t, H5 implies that upward revisions in expected future volatility should tend to occur following unscheduled announcements. In fact, Brown, Harlow, and Tinic (1993) finds that following large price changes (which the study attributes to information events) actual volatility increases in some cases and decreases in some, but increases dominate. Consequently, we hypothesize,

H6. The IMV will tend to rise following important unscheduled announcements that cause volatility on day t to be higher than anticipated.

In a related study, French, Swidler, and Trapani (1994) finds approximately equal increases and decreases in implied volatility following 8-K filings—an unscheduled release. Nonetheless, our theory indicates that IMV increases should dominate following most important unscheduled releases.

In summary, we argue that for *scheduled* announcements, the high volatility on day t is generally foreseen and the ISD declines as this uncertainty is resolved. In the case of *unscheduled* announcements, we argue that the high day t volatility is not foreseen and the ISD normally rises because the unanticipated high volatility on day t usually causes market participants to revise upward their expectations of likely volatility over the remaining life of the option.

Note that expected and unexpected (actual-expected) volatility are hypothesized to have opposite effects on the ISD. If *expected* volatility on day t is high (low), the first bracketed term in equation (6) will be negative (positive) and the ISD will tend to fall (rise) since this day will no longer be covered by the option, H1 and H2. If *actual* volatility is higher (lower) than expected, the second bracketed term will be positive (negative) and the ISD will tend to rise (fall), H5. Finally, the high actual volatility caused by unscheduled announcements will increase the ISD, H6.

B. Testing the Impact of Unscheduled Announcements

Testing H6 is complicated by the fact that, due to their nature, we cannot identify specific unscheduled announcements. The markets receive several news items every day and identifying the specific news release responsible for the price volatility on day t is extremely difficult. Assuming, however, that large price innovations are due to some sort of information release, most large price changes on days with no scheduled announcements can probably be attributed to an unsched-

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

uled announcement that day,[15] while large price changes on days with scheduled announcements could be due to either the scheduled announcement or to an unscheduled announcement that also happened to occur that day. Consequently, to test H6, we focus on days without scheduled announcements, which complicates our test slightly. Note that while we argue in Section III.A that for unscheduled announcements, in general (that is, including those that occur on days with scheduled announcements), the first bracketed term in equation (6) averages to zero, for this subsample of days without scheduled announcements, a positive first bracketed term is expected due to H2.

Our measure of actual volatility on day t is the absolute price innovation, $|R_t|$, where $R_t = \ln(F_t/F_{t-1})$ and F_t is the closing price of the underlying futures contract on day t. To test H5 and H6, we regress the percentage change in the ISD on the percentage difference between actual and expected volatility on day t,

$$(10) \qquad \ln\left(\sigma_t/\sigma_{t-1}\right) = \left(\frac{1}{2T_t}\right)\left[\gamma_0 + \gamma_1\left(\frac{|R_t| - \hat{E}_{t-1}|R_t|}{\hat{E}_{t-1}|R_t|}\right)\right] + v_t,$$

over the sample of days without scheduled announcements.[16] Comparing the empirical equation (10) with the theoretical equation (6), γ_0 in (10) estimates the first bracketed term in (6), i.e., the deviations of expected day t volatility from mean expected volatility. Since we are estimating (10) over the sample of days without scheduled announcements, H2 implies $\gamma_0 > 0$. Likewise, $\gamma_1[(|R_t| - \hat{E}_{t-1}|R_t|)/\hat{E}_{t-1}|R_t|]$ provides an estimate of the second bracketed term in (6), and H5 and H6 imply $\gamma_1 > 0$. For $\hat{E}_{t-1}|R_t|$, we use the mean of $|R_t|$ on days without scheduled announcements: 0.003546 for T-Bonds and 0.000360 for Eurodollars.

Results of the OLS estimation of equation (10) over the sample of days without scheduled announcements are shown in the first column (labeled Model 1) for each market in Table 6 . Reconfirming H2, $\hat{\gamma}_0$ is positive and significant. More importantly, consistent with H5 and H6, $\hat{\gamma}_1$ is positive and significant at the 0.01 level in both the T-Bond and Eurodollar markets, implying that following major unscheduled announcements (and the higher than expected volatility that they generate), market participants tend to revise their expectations of future volatilities upward causing a rise in the ISD.

While we expect $\hat{E}_{t-1}|R_t|$ to be roughly the same for all days without scheduled announcements, it should vary across days with scheduled announcements depending on the normal impact of the announcement on volatility. In other words, $\hat{E}_{t-1}|R_t|$ should be higher if t is a day the employment report is scheduled to be released than if the retail sales report is scheduled for release. However, for comparison with the estimation over days without scheduled announcement, we first estimate equation (10), employing the same procedure of using mean values for

[15] In our sample, it is of course possible that some large price innovations on days without scheduled announcements are due to scheduled announcements that we fail to consider, such as a scheduled speech by the Federal Reserve Board chairman.

[16] While the independent and dependent variables are both percentages, the former is expressed in algebraic form and the latter in log form. This is because we have some observations for which $|R_t|$ is zero, so that the log of $|R_t|$ is undefined.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

TABLE 6

The Impact of Expected and Unexpected Volatility on the Implied Standard Deviation Following Scheduled and Unscheduled Announcements

	Coefficient Estimates with Standard Errors (in parentheses)							
	T-Bond Regressions				Eurodollar Regressions			
Coefficient	(1) Without Sch. Ann.	(2) With Sch. Ann.	(3) With Sch. Ann.	(4) All Days	(1) Without Sch. Ann.	(2) With Sch. Ann.	(3) With Sch. Ann.	(4) All Days
γ_0	0.244** (0.063)	−0.313** (0.093)		0.266** (0.054)	0.403** (0.082)	−0.716** (0.121)		0.349** (0.075)
γ_1	0.163** (0.063)	−0.146* (0.080)	0.113 (0.100)	0.191** (0.062)	0.314** (0.076)	0.056 (0.099)	0.305 (0.135)	0.284** (0.078)
γ_2				−0.036 (0.111)				0.010 (0.149)
Adj. (Total) R^2	0.034	0.045	0.161	0.147	0.060	0.077	0.176	0.142

Estimates of the following regressions are reported,

$$\ln\left(\sigma_t/\sigma_{t-1}\right) = \left(\frac{1}{2T_t}\right)\left[\gamma_0 + \sum_{k=1}^{K}\beta_k D_{kt} + \gamma_1\left(\frac{|R_t| - \hat{E}_{t-1}|R_t|}{\hat{E}_{t-1}|R_t|}\right) + \gamma_2 D_t\left(\frac{|R_t| - \hat{E}_{t-1}|R_t|}{\hat{E}_{t-1}|R_t|}\right)\right] + v_t,$$

where

σ_t = the implied standard deviation on day t as calculated from daily closing futures and option prices and adjusted to a market day basis,
$|R_t|$ = the absolute value of $\ln(F_t/F_{t-1})$ where F_t is the closing futures price on day t—a measure of actual volatility,
$E_{t-1}|R_t|$ = the expected (as of day $t-1$) volatility on day t,
T_t = the time to expiration of the option contract,
D_t = 1 if a scheduled announcement occurs on day t and 0 otherwise, and
D_{kt} = 1 if scheduled announcement k occurs on day t and 0 otherwise.

Model 1: sample = days without scheduled announcements (so D_t and $D_{kt} = 0$),
$\hat{E}_{t-1}|R_t|$ = mean of $|R_t|$ on days without scheduled announcements.
Model 2: sample = days with scheduled announcements, β_k (all k) and $\gamma_2 = 0$,
$\hat{E}_{t-1}|R_t|$ = mean of $|R_t|$ on days with scheduled announcements.
Model 3: sample = days with scheduled announcements, γ_0 and $\gamma_2 = 0$,
$\hat{E}_{t-1}|R_t| = \hat{\alpha}_0 + \sum_{k=1}^{K}\hat{\alpha}_k D_{kt}$ where the $\hat{\alpha}$s are from Table 1.
Model 4: sample = complete sample, no parameter restrictions,
$\hat{E}_{t-1}|R_t| = \hat{\alpha}_0 + \sum_{k=1}^{K}\hat{\alpha}_k D_{kt}$ where the $\hat{\alpha}$s are from Table 1.

The regressions are estimated using daily data from 11/11/88 through 9/30/92. * and ** denote parameters that are significantly different from zero at the 0.05 and 0.01 levels in one-tail tests, respectively.

$\widehat{E}_{t-1}|R_t|$—in particular the mean of $|R_t|$ on days with scheduled announcements: 0.004978 for T-Bonds and 0.000606 for Eurodollars. Results are shown in column 2 (Model 2) for each market in Table 6. As implied by H1, $\widehat{\gamma}_0$ is negative and significant. However, $\widehat{\gamma}_1$ is small and insignificant for Eurodollars, and is even negative in the T-Bond market. As demonstrated below, the insignificance of the γ_1 coefficients is due to the fact that Model 2 does not control for variations in expected volatility across days with scheduled announcements.

C. Expected and Unexpected Day t Volatility and the ISD

While H6 pertains to unscheduled announcements, H5 is more general, so it is important to test whether it holds for days with scheduled announcements as well. To test H5 for days with scheduled announcements, we allow expected volatility to vary across announcements. First, we calculate $(|R_t| - \widehat{E}_{t-1}|R_t|)/\widehat{E}_{t-1}|R_t|$, using estimates of $\widehat{E}_{t-1}|R_t|$ obtained from the estimates of equation (7) in Table 1, specifically $\widehat{E}_{t-1}|R_t| = \widehat{\alpha}_0 + \sum_{k=1}^{K}\widehat{\alpha}_k D_{kt}$, where $D_{kt} = 1$ if announcement k is

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

released on day t and the $\hat{\alpha}$s are from Table 1. In other words, $\hat{\alpha}_0$ is our estimate of $\hat{E}_{t-1}|R_t|$ on days without scheduled announcements, and $\hat{\alpha}_0 + \hat{\alpha}_k$ is our estimate of $\hat{E}_{t-1}|R_t|$ on days with scheduled announcement k. The resulting $|R_t| - \hat{E}_{t-1}|R_t|$ is our measure of unexpected volatility on day t. Second, to control for the impact of fluctuations in *expected* volatility on the ISD, we replace γ_0 in equation (10) with separate intercepts for each announcement type k. We then estimate the equation,

$$(11) \quad \ln(\sigma_t/\sigma_{t-1}) = \left(\frac{1}{2T_t}\right)\left[\gamma_0 + \sum_{k=1}^{K}\beta_k D_{kt} + \gamma_1\left(\frac{|R_t| - \hat{E}_{t-1}|R_t|}{\hat{E}_{t-1}|R_t|}\right) + \gamma_2 D_t\left(\frac{|R_t| - \hat{E}_{t-1}|R_t|}{\hat{E}_{t-1}|R_t|}\right)\right] + v_t,$$

over the entire sample of days with and without scheduled announcements. $D_t = 1$ if there is a scheduled announcement on day t, and $D_t = 0$ otherwise. Note that on days without scheduled announcements, equation (11) reduces to equation (10). Also, γ_0 and $\gamma_0 + \beta_k$ provide estimates of the first bracketed term in equation (6) on days without scheduled announcements and on days with announcement k, respectively. H2 implies $\gamma_0 > 0$ and H1 implies $\gamma_0 + \beta_k < 0$ for all k. The term $\gamma_1[(|R_t| - \hat{E}_{t-1}|R_t|)/\hat{E}_{t-1}|R_t|]$ provides an estimate of the second term in equation (6) on days without scheduled announcements, while $(\gamma_1 + \gamma_2)[(|R_t| - \hat{E}_{t-1}|R_t|)/\hat{E}_{t-1}|R_t|]$ provides an estimate of the second term on days with scheduled announcements. H6 implies $\gamma_1 > 0$, while H5 implies both $\gamma_1 > 0$ and $(\gamma_1 + \gamma_2) > 0$. The coefficient γ_2 measures any difference between the impact of unexpected volatility on the ISD on days with and without scheduled announcements.

Results are reported in column 3 (Model 3) in each panel of Table 6 . Since the entire sample, unlike those used to estimate Models 1 and 2, is continuous, in estimating Model 3, we can correct for the autocorrelation in the error terms, yielding more efficient estimates than those in Models 1 and 2. Reconfirming H2, $\hat{\gamma}_0$ is positive and significant at the 0.01 level in both markets. As predicted by H1, most of the estimated βs (which are not shown due to space limitations but are available on request) are negative. Together, these results indicate that if expected volatility is high (low) on day t, the ISD tends to fall (rise). Confirming H6, $\hat{\gamma}_1$ is positive and significant at the 0.01 level in both markets and, confirming H5, $(\hat{\gamma}_1 + \hat{\gamma}_2) > 0$. In the T-Bond market, $\hat{\gamma}_2$ is negative, small, and insignificant while, in the Eurodollar market, $\hat{\gamma}_2$ is positive, small, and insignificant. Overall, these results imply that high (low) unexpected volatility on day t causes the ISD to rise (fall) and this impact is approximately the same whether the unexpected volatility is due to a scheduled or unscheduled announcement.

It is interesting to compare our results with ARCH-GARCH models and other work on volatility persistence. In ARCH-GARCH models, higher than expected volatility on day t implies an upward revision in expected volatility on day $t + 1$ and later, and Engle and Mustafa (1992) argues that the same pattern should be observed in implied volatilities. Our evidence documents that ISDs vary directly with unexpected day t volatility, but also indicates: i) that ISDs vary negatively with expected volatility, and ii) that, unlike ARCH-GARCH models that base expected volatilities solely on recent historical volatilities, market participants utilize knowl-

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

edge of the timing of scheduled news releases in forming their expectations. Day and Lewis (1992) and Lamoureux and Lastrapes (1993) find that adding the ISD to a GARCH (1,1) model improves its ability to predict future volatility. Our results suggest one factor that market participants consider, but that present GARCH models do not, is whether the price shock is due to a scheduled or unscheduled information release.

IV. Market Efficiency

Attention is now turned to market efficiency. A number of studies, including Canina and Figlewski (1993), Lamoureux and Lastrapes (1993), and Day and Lewis (1992) have questioned whether implied volatilities from options markets fully incorporate all available information. For instance, Canina and Figlewski (1993) finds that historical volatilities yield more accurate forecasts of future volatilities than implied volatilities. Our results are relevant to this debate because we find that market participants apparently do take into account the schedule of upcoming announcements in forming their expectations of likely future volatility.

However, our results raise a separate efficiency issue. To wit, if the ISD consistently falls following major releases like the employment report, is it possible to construct a profitable trading strategy based on this tendency? After finding that daily ISDs are partially predictable based on lagged volatility changes and Monday and Friday dummies, Harvey and Whaley (1992) investigates whether abnormal trading profits are possible and find that they are not because of transaction costs. However, the ISD declines following scheduled releases, which we document are much larger than the weekday patterns examined in Harvey and Whaley and would appear to dwarf transaction costs. For instance, in our Eurodollar data set, the mean ISD decline following the release of the employment report is more than 20 times the mean Friday to Monday ISD change considered in Harvey and Whaley.

For this analysis, we focus on the report for which possible trading profits would appear greatest—the employment report, which causes the ISD to decline more than 80 percent of the time in our sample. In our T-Bond sample, the ISD averages 0.1159 before the release of the employment report and 0.1111 after; F_{t-1} averages 95.97 and T_{t-1} averages 32.6 market days. If we consider a call with these attributes and a strike price approximately equal to F_{t-1}, this decline in σ translates into a decline in the call price of \$88.67.[17] If we estimate transaction costs on both the purchase and sale at one tick, the profits net of transaction costs if one writes one call just prior to the release of the employment report and buys it back after would be \$88.67 − \$31.25 = \$57.42.

This example assumes no change in the underlying futures price F_t. This strategy of writing a call would expose one to a loss risk if the underlying futures price should rise, which it could due to the employment report release. Since $\partial C_t/\partial F_t = e^{-rT}N(d_1)$ and $\partial P_t/\partial F_t = -e^{-rT}N(-d_1)$ (where P_t is the put price and the other terms are as defined in Section II.A), the usual prescription for hedging against such price change risk is to form a *delta neutral* portfolio in which one

[17] The quoted call price declines \$0.08867. Since the T-Bond futures and call are for \$100,000 face value of Treasury Bonds, while the call is quoted on a \$100 face value basis, the decline in the price of one call is \$88.67.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

writes $N(d_1)/N(-d_1)$ puts for each call written. For such a delta neutral portfolio, the loss (profit) on the call due to a *small* rise (fall) in F_t is offset by a profit (loss) on the puts. Since both the call and put price are directly related to σ, profits are realized on both the call and put positions if the ISD falls.

This is illustrated in Table 7. If the futures price, F, is unchanged, then, as shown in Table 7, the profits before transaction costs on the put position in this hypothetical transaction would be \$91.71 so total profits before transaction cost on this portfolio would be \$180.38 before transaction costs and \$116.80 after transaction costs. However, as shown in Table 1, release of the employment report often causes large changes in F. Specifically, on days the employment report is released, the standard deviation of $F_t - F_{t-1}$ is \$0.87134 per \$100 of face value. Suppose that σ falls from 0.1159 to 0.1111 as before, but suppose the underlying futures price also rises. Suppose first that the rise in F is relatively small, e.g., by half of one standard deviation of the normal change, i.e., by \$0.87134/2 from F_{t-1} = \$95.97 to F_t = \$96.40. As shown in Table 7, in this case, there is pretransaction-cost loss of \$141.28 on the call but a profit of \$302.17 on the put position so total profits are \$160.89, or approximately the same as in the no futures price change case. In the case of this small price change, the hedge works. As shown in Table 7, an equal fall in F results in approximately the same total profits though the profitability positions of the call and put are reversed.

TABLE 7

Simulated Profits from Writing a Delta Hedged Call-Put Portfolio Prior to Release of the Employment Report and Longing Post-Release—Based on Mean Option Parameters

	Change in the Underlying Futures Price, F, Expressed as a Proportion of the Standard Deviation of the Price Change on Days the Employment Report is Released, S								
Position	No Chg.	$+0.5S$	$-0.5S$	$+S$	$-S$	$+1.5S$	$-1.5S$	$+2S$	$-2S$
Panel A. T-Bond Market Profits or Losses									
Short Call	\$88.67	\$−141.28	\$299.00	\$−390.58	\$489.73	\$−658.73	\$661.08	\$−945.04	\$813.56
Short Put	91.71	302.17	−139.03	492.61	−390.05	663.54	−661.11	815.71	−951.69
Total	\$180.38	\$160.89	\$159.98	\$102.03	\$99.67	\$4.81	\$−0.03	\$−129.33	\$−138.13
After Trans. Costs	\$116.80	\$97.31	\$96.40	\$38.45	\$36.09	\$−58.77	\$−63.61	\$−192.91	\$−201.71
Panel B. Eurodollar Market Profits or Losses									
Call	\$48.79	\$−40.89	\$129.71	\$−139.23	\$201.95	\$−246.02	\$265.70	\$−360.90	\$321.30
Put	49.03	129.97	−40.71	202.20	−139.17	265.95	−246.17	321.55	−361.35
Total	\$97.82	\$89.09	\$89.01	\$62.97	\$62.78	\$19.92	\$19.53	\$−39.36	\$−40.04
After Trans. Costs	\$47.69	\$38.96	\$38.88	\$12.84	\$12.65	\$−30.21	\$−30.60	\$−84.49	\$−90.17

Shown are the simulated profits to writing one call and $N(d_1)/N(-d_1)$ puts on day $t-1$ and buying both back on day t. Transaction costs are assumed to be one tick per contract on both the purchase and sale. The strike price is assumed equal to F_{t-1}.

In the T-Bond market, it is assumed F_{t-1} = 95.966, σ_{t-1} = 0.11587, σ_t = 0.11115, T_t = 32.61. These are the mean values in the T-Bond market on the Fridays when the employment report is released over the period 11/88–9/92. The standard deviation of $(F_t - F_{t-1}) = S$ = 0.87135 and $N(d_1)/N(-d_1)$ = 1.03434.

In the Eurodollar market, it is assumed F_{t-1} = 92.878, σ_{t-1} = 0.01807, σ_t = 0.01682, T_t = 29.24. These are the mean values in the Eurodollar market on the Fridays when the employment report is released over the period 11/88–9/92. The standard deviation of $(F_t - F_{t-1}) = S$ = 0.13732 and $N(d_1)/N(-d_1)$ = 1.00501.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

Unfortunately, while this portfolio is hedged against small changes in the underlying futures price, it is impossible to hedge against large price changes and the convexity of option prices means that even on a delta neutral portfolio, large price changes will tend to create losses. While the first derivatives of call and put prices with respect to F_t are of opposite sign so that it is possible to construct a delta neutral portfolio for which the first derivative is zero, both calls and puts have positive second derivatives or *gammas*. Since both calls and puts are shorted in the delta neutral trading portfolio, its gamma is negative. This means that, if there is a *large* increase in F_t, the loss on the short call position will tend to exceed the profit on the short put. If there is a large drop in F_t, the loss on the put exceeds the profit on the call. This is also illustrated in Table 7. Suppose that σ falls from 0.1159 to 0.1111 as before, but suppose the underlying futures price rises by two standard deviations, i.e., from F_{t-1} = \$95.97 to F_t = \$97.71. As shown in Table 7, in this case, there is a pretransaction-cost loss of \$945.04 on the call and a profit of \$815.70 on the put position, yielding a total loss of \$129.33 before transaction costs. In summary, it is possible to hedge against small futures price changes caused by the release of the employment report but not against large changes. The profits and losses on this hypothetical portfolio are shown in Table 7 for various possible changes in the futures price. If the futures price changed gradually over time, it would be possible to hedge against large changes by rebalancing the portfolio but since the main adjustment to the employment report occurs within one minute (Ederington and Lee (1995)), that is not possible here.

In Table 8, we report profit statistics for our sample on a strategy of writing one call and $N(d_1)/N(-d_1)$ puts on the Thursday prior to the employment report releases (at the settlement prices) and buying back the next day after the release (at Friday's settlement price). Since it is easily shown that $\partial C_t/\partial\sigma$ is maximized where the exercise price is close to but slightly above F_t, we construct delta neutral trading portfolios based on the first out-of-the-money call and its twin put. For comparison, we also examine portfolios based on the first in-the-money call and its twin put. Pretransaction-cost profits on the out-of-the-money portfolios average only \$36.25 in the T-Bond market and \$8.39 in the Eurodollar-profits that would not cover one tick transaction costs.

As demonstrated in Panels B and C of Table 8, these low mean and median profits are due to the aforementioned convexity of option prices, which means that, while our trading portfolio is hedged against small changes in F_t, large changes cause losses. To explore this, we divide our 46 employment report releases into the 23 days, Panel B, with the smallest ex post values of $|\ln(F_t/F_{t-1})|$ and the 23 days with the largest values, Panel C. As shown in Panel B, when the absolute change in the futures price is small, the trading strategy works fairly well. However, when the absolute change in the futures price is large, large losses tend to be observed as shown in Panel C. In summary, on average, the profits from the ISD decline are offset by losses due to the high actual volatility occasioned by the release and the convexity of option prices.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

TABLE 8

Pretransaction Cost Returns to a Trading Strategy Based on the Predictable ISD Declines following Release of the Employment Report

	Mean	Median	% Profitable	Std. Dev.
Panel A. All 46 Employment Releases				
T-Bond:				
Out-of-the-Money Call	$36.25	$76.54*	65%	$163.87
In-the-Money Call	$36.85	$83.93	65%	$260.92
Eurodollar:				
Out-of-the-Money Call	$8.39	$29.51*	70%**	$90.58
In-the-Money Call	$−42.49	$15.38	54%	$185.98
Panel B. 23 Employment Releases with the Smallest Absolute Impact on the Futures Price				
T-Bond:				
Out-of-the-Money Call	$114.55**	$92.15**	91%**	$90.80
In-the-Money Call	$178.34**	$169.13**	91%**	$142.75
Eurodollar:				
Out-of-the-Money Call	$56.81**	$50.00**	96%**	$42.71
In-the-Money Call	$62.91**	$50.62**	78%**	$84.09
Panel C. 23 Employment Releases with the Largest Absolute Impact on the Futures Price				
T-Bond:				
Out-of-the-Money Call	$−42.05	$−56.99	39%	$184.02
In-the-Money Call	$−104.64	$−80.89	39%	$277.52
Eurodollar:				
Out-of-the Money Call	$−40.02	$−10.59	43%	$100.28
In-the-Money Call	$−147.89**	$−157.97**	30%*	$201.12

Results for a trading strategy of writing one call and $N(d_1)/N(-d_1)$ puts (with the same exercise price) on the day before the employment report release and buying the same options at their closing price on the day of the release. The ratio of puts to calls is chosen to hedge against small changes in F_t. The reported profits are for one call contract and $N(d_1)/N(-d_1)$ puts. Separate strategies based on the first out-of-the-money call and first in-the-money call are reported. * and ** denote significance at the 0.05 and 0.01 levels, respectively, in two-tailed tests. For the mean (median), the null hypothesis is that $\mu(M) = 0$; for the percent profitable, it is $p = 0.5$.

V. Summary

We have explored the determinants of market participants' uncertainty regarding future security prices as measured by the ISD. We find that scheduled announcements generally lead to a drop in the ISD as the uncertainty regarding the impact of the announcement on security prices is resolved. Conversely, most major unscheduled announcements cause an increase in market uncertainty and a rise in the ISD. Looked at another way, our results indicate that high expected volatility on day t leads to a fall in the ISD as this uncertainty is resolved while high unexpected volatility on day t leads to a rise in the ISD as market participants revise upward their estimates of likely volatility over the remaining life of the option. Our results have implications for researchers of volatility persistence. While ARCH and other models of volatility persistence imply an increase in the conditional variance following large price innovations, we find that the ISD only rises following price innovations due to unscheduled announcements. It normally declines following price innovations due to scheduled announcements.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

APPENDIX

Macroeconomic Announcements

Time of Release	Short Title	Full Title of Report	Reporting Agency	Day-of-the-Week M	T	W	R	F
8:30 am	Consumer Price Index	Consumer Price Index	Bureau of Labor Statistics	0	16	11	10	10
8:30 am	Durable Goods Orders	Advance Report on Durable Goods Manufacturers' Shipments and Orders	Bureau of the Census	0	14	16	9	8
8:30 am	Employment	The Employment Situation	Bureau of Labor Statistics	0	0	0	1	45
8:30 am	Gross National Product	Gross National Product	Bureau of Economic Analysis	0	8	14	14	10
8:30 am	Merchandise Trade Deficit	Advance Report of U.S. Merchandise Trade Deficit	Bureau of the Census	0	7	11	15	14
8:30 am	Producer Price Index	Producer Price Indexes	Bureau of Labor Statistics	0	1	2	9	33
8:30 am	Retail Sales	Advance Retail Sales	Bureau of the Census	0	16	6	13	12
9:15 am	Industrial Production	Industrial Production	Federal Reserve Board	2	13	10	4	18
9:15 am*	Capacity Utilization	Capacity Utilization	Federal Reserve Board	2	12	11	4	18
10:00 am	Construction Spending	Value of New Construction Put in Place	Bureau of the Census	16	8	8	7	7
10:00 am	NAPM Survey	NAPM Survey	National Association of Purchasing Managers	20	8	6	5	7
2:00 pm	Federal Budget	Treasury Statement (monthly budget)	Department of the Treasury	10	8	9	6	14
			Total:	50	111	104	97	196

*In two months the announcement was made at 10:30 and in one its timing was unclear.

References

Bailey, W. "Money Supply Announcements and the Ex-Ante Volatility of Asset Prices." *Journal of Money, Credit and Banking*, 20 (1988), 611–620.

Ball, C., and A. Roma. "Stochastic Volatility Option Pricing." *Journal of Financial and Quantitative Analysis*, 29 (1994), 589–607.

Black, F. "The Pricing of Commodity Contracts." *Journal of Financial Economics*, 3 (1976), 167–179

Bollerslev, T.; R. Y. Chou; and K. F. Kroner. "ARCH Modeling in Finance: A Review of the Theory and Empirical Evidence." *Journal of Econometrics*, 52 (1992), 5–59.

Bonser-Neal, C., and G. Tanner. "Central Bank Intervention and the Volatility of Foreign Exchange Rates: Evidence from the Options Market." Working Paper #95-04, Federal Reserve Bank of Kansas City Research (1994).

Brown, K.; W. Harlow; and S. Tinic. "The Risk and Return of Common Stock following Major Price Innovations." *Journal of Financial and Quantitative Analysis*, 28 (1993), 101–116.

Butler, J., and B. Schachter. "Unbiased Estimation of the Black/Scholes Formula." *Journal of Financial Economics*, 15 (1986), 341–357.

Canina, L., and S. Figlewski. "The Informational Content of Implied Volatility." *Review of Financial Studies*, 6 (1993), 659–681.

Chiras, D., and S. Manaster. "The Informational Content of Option Prices and a Test of Market Efficiency." *Journal of Financial Economics*, 6 (1978), 213–234.

Choi, S., and M. E. Wohar. "Implied Volatility in Options Markets and Conditional Heteroskedasticity in Stock Markets." *Financial Review*, 27 (1992), 503–530.

Cornell, B. "Using the Option Pricing Model to Measure the Uncertainty Producing Effect of Major Announcements." *Financial Management*, 7 (1978), 54–59.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press

Day, T. E., and C. M. Lewis. "The Behavior of the Volatility Implicit in the Prices of Stock Index Options." *Journal of Financial Economics*, 22 (1988), 103–122.

_____. "Stock Market Volatility and the Information Content of Stock Index Options." *Journal of Econometrics*, 52 (1992), 267–287.

Ederington, L. H., and J. H. Lee. "How Markets Process Information: News Releases and Volatility." *Journal of Finance*, 48 (1993), 1161–1191.

_____. "The Short-Run Dynamics of the Price Adjustment to New Information." *Journal of Financial and Quantitative Analysis*, 30 (1995), 117–134.

Engle, R. F., and C. Mustafa. "Implied ARCH Models from Options Prices." *Journal of Econometrics*, 52 (1992), 289–311.

Franks, J. R., and E. S. Schwartz. "The Stochastic Behavior of Market Variance Implied in the Prices of Index Options." *Economic Journal*, 101 (1991), 1460–1475.

French, D. W. "The Weekend Effect on the Distribution of Stock Prices." *Journal of Financial Economics*, 13 (1984), 547–559.

French, D.W., and D. Frasier. "A Note on Interest Rates and the Risk of Bank and Savings and Loan Stock." *Financial Review*, 21 (1986), 551–558.

French, D.; S. Swidler; and T. Trapani. "Disclosure of Corporate Events: An Examination of the SEC's 8-K Report. Working Paper, Univ. of Texas at Arlington (1994).

Harvey, C. P., and R. E. Whaley. "Market Volatility Prediction and the Efficiency of the S&P 100 Index Option Market. *Journal of Financial Economics*, 31 (1992), 43–74.

Heynen, R.; K. Kemna; and T. Vorst. "Analysis of the Term Structure of Implied Volatilities." *Journal of Financial and Quantitative Analysis*, 29 (1994), 31–56.

Hull, J., and A. White. "The Pricing of Options on Assets with Stochastic Volatilities." *Journal of Finance*, 42 (1987), 281–300.

Jorion, P. "Predicting Volatility in the Foreign Exchange Market." *Journal of Finance*, 50 (1995), 507–528.

Kim, O.; and R. E. Verrecchia. "Market Reaction to Anticipated Announcements." *Journal of Financial Economics*, 30 (1991), 273–310.

Lamoureux, C. G., and W. D. Lastrapes. "Forecasting Stock-Return Variance: Toward an Understanding of Stochastic Implied Volatilities." *Review of Financial Studies*, 6 (1993), 293–326.

Latané, H. A., and R. J. Rendleman, Jr. "Standard Deviations of Stock Price Ratios Implied in Options Prices." *Journal of Finance*, 31 (1976), 369–381.

Levy, H., and J. Yoder. "The Behavior of Option Implied Standard Deviations around Merger and Acquisition Announcements." *Financial Review*, 28 (1993), 261–272.

Merton, R. "Theory of Rational Option Pricing." *Bell Journal of Economics and Management Science*, 4 (1973), 141–183.

Merville, L. J., and D. R. Pieptea. "Stock-Price Volatility, Mean-Reverting Diffusion, and Noise." *Journal of Financial Economics*, 24 (1989), 193–214.

Patell, J. M., and M. A. Wolfson. "Anticipated Information Releases Reflected in Call Option Prices." *Journal of Accounting and Economics*, 1 (1979), 117–140.

_____. "The Ex Ante and Ex Post Price Effect of Quarterly Earnings Announcements Reflected in Option and Stock Prices." *Journal of Accounting Research*, 19 (1981), 434–458.

Xu, X., and S. Taylor. "Conditional Volatility and the Informational Efficiency of the PHLX Currency Options Market." *Journal of Banking and Finance*, 19 (1995), 803–822.

https://doi.org/10.2307/2331358 Published online by Cambridge University Press